IOT ENABLED 3-PHASE POWER QUALITY & ENERGY METER

TUE-17

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RESPONSE TO FEEDBACK FROM MILESTONE-2

1. **FEEDBACK:** Work on documentation

   RESPONSE: We have worked and clearly documented our work and findings on One-Note folder in Teams

TESTING RESULTS

1. Demo data correctly sent to server and Mobile is correctly able to fetch the same demo data of energy/voltage/current
2. The ADE9000 Evaluation Board was tested with a load, and the data sent to the server was transmitted to a mobile app via the server using a **single-phase load**
3. Established relay (Remotely On-Off) using Web Socket and checked it on LED.
4. In case of failing of data being sent to server, we are able to store data on local database

Issues: No deviations from the expected result were found as of now.

PROOF OF CONCEPT

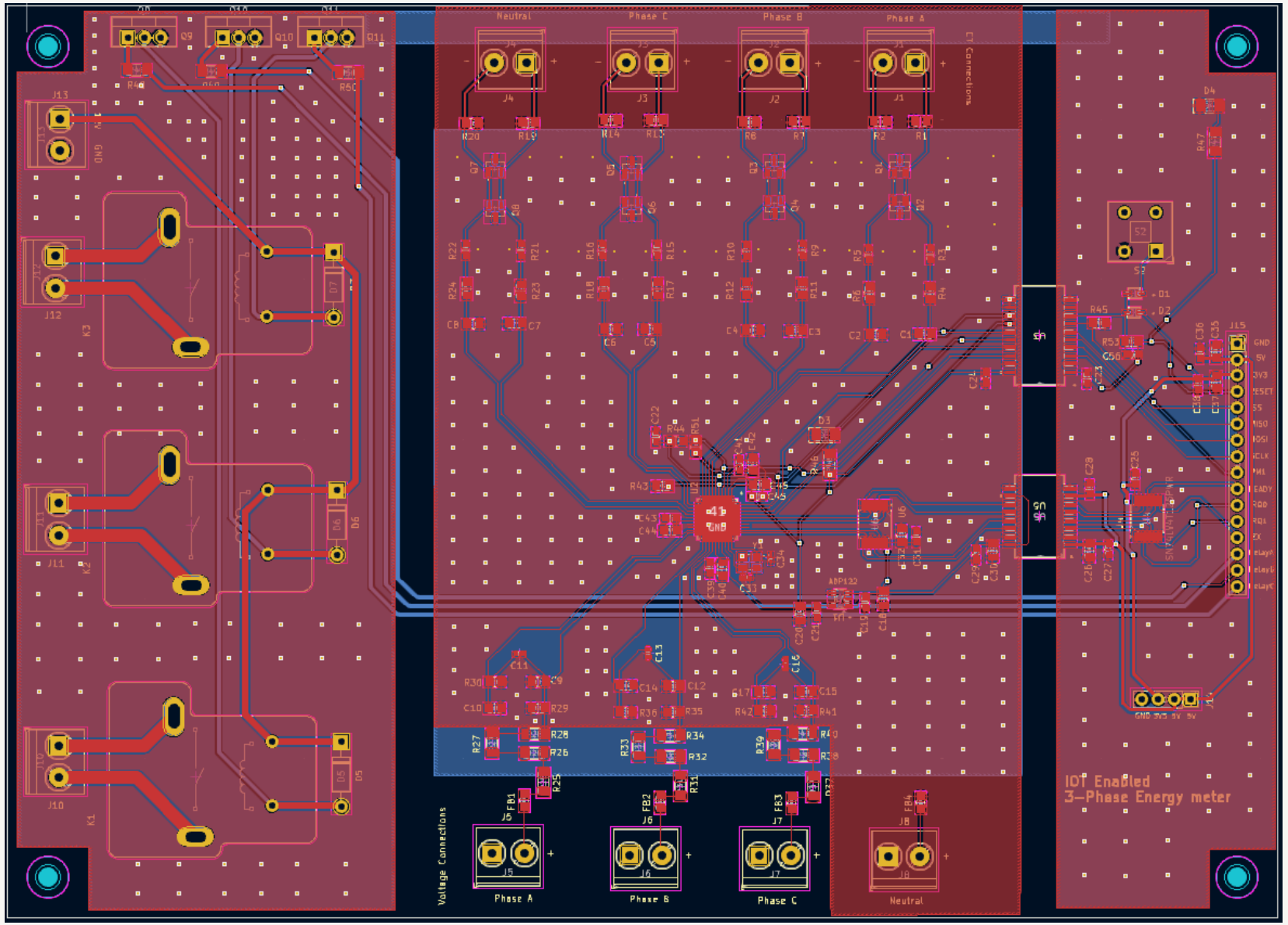
Video Link 1: Connection check between ADE9000, RPI and Android App

Video Link 2: Relay Check

PCB READINESS

Our PCB layout is completed and verified by Maheshwar sir and was ordered on 11th March. Payment was also done on 21st.
It is expected to come any day now.

PCB LAYOUT

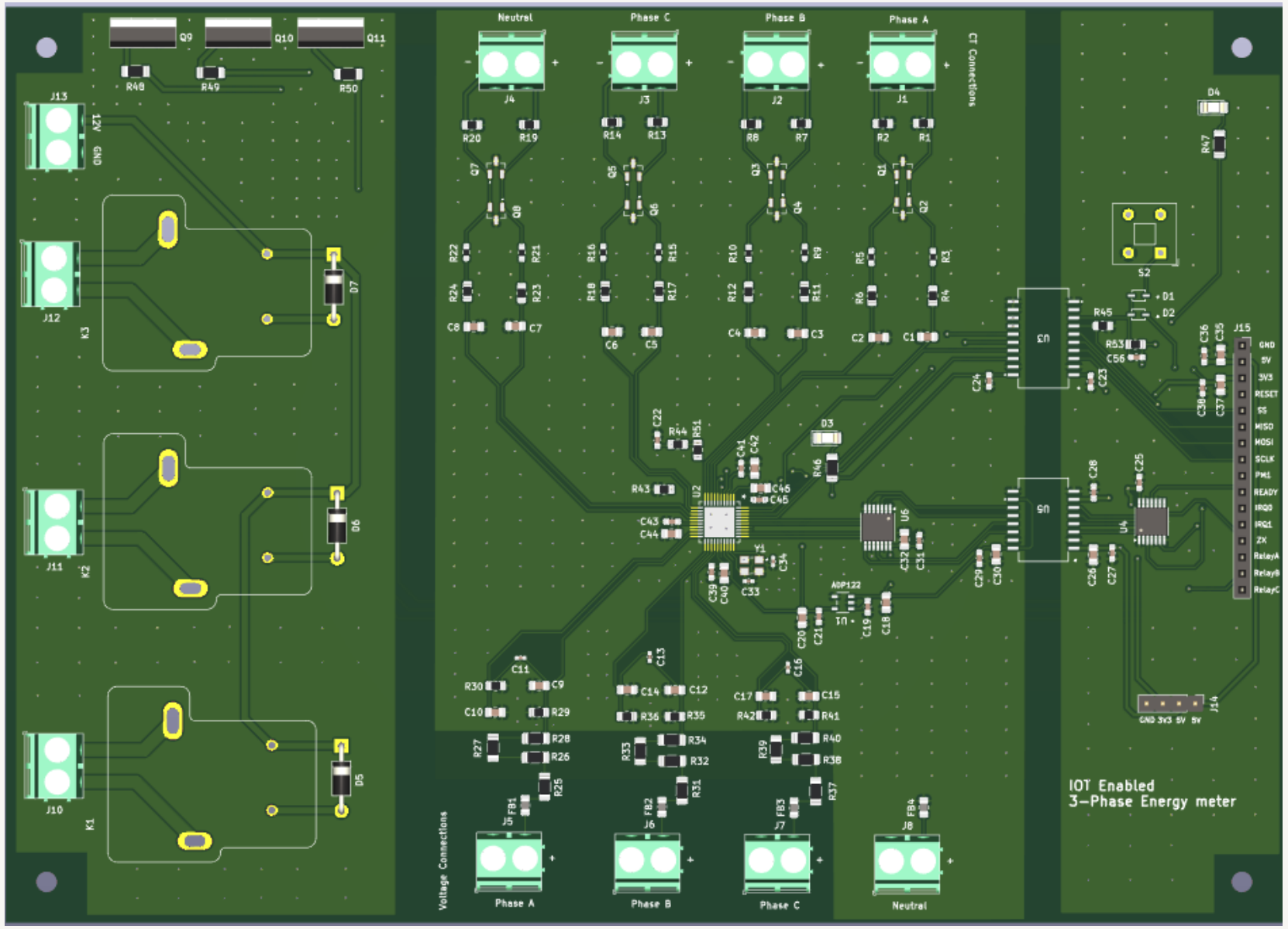

Neutral
Phase C
Phase B
Phase A
CT Connections
Voltage Connections
Phase A
Phase B
Phase C
Neutral
IOT Enabled
3-Phase Energy meter

PCB DESIGN CHOICES AND KEY LEARNINGS

Based on multiple review feedback from WEL lab RAs and staff, we incorporated these key design considerations:

- Isolated ground planes for high and low voltage sections
- Placed stitching vias everywhere for better grounding
- Positioned headers at the edges for easier connectivity
- Placed relay drive transistors at the edge for heat sink placement
- Used appropriate trace widths for different traces based on current ratings
- Added extra solder mask for ADE9000 IC to ease soldering
- Used equal trace length for differential signal routing
- Placed decoupling capacitors closed to IC pins wherever required

PCB 3D VIEW


Neutral
Phase C
Phase B
Phase A
CT Connections
Voltage Connections
IOT Enabled
3-Phase Energy meter

CRITICAL TASKS & KEY RISKS

1. Final Testing on PCB is remaining, all the individual tasks are completely tested on ADE9000 Evaluation Board.
2. After completing soldering and verification on the PCB, we need to order the LEGRAND box for final casing to ensure insulation and seamless, risk-free operation. This can only be done after the PCB arrives, as the box needs to be cut to accommodate the PCB and LCD screen as per the final size of PCB.
3. Check the final robustness of system with 3-phase Load

PROGRESS AGAINST ORIGINAL TEST PLAN

1. Everything has been done as per the Original Gant chart in Milestone except that we had anticipated that PCB will come in the week of 11th March and we will be done by soldering at the end of this week.
2. But since PCB will come next week, we haven't completed that as was written in our original Gantt chart.
3. Rest everything has been completed

WORK DISTRIBUTION UNTIL FINAL DEMO

1. **Saurabh** - Testing of Server with RPI and Connection between Android app and server
2. **Sarthak** - Working on UI for LCD Screen
3. **Aryan** - Soldering on PCB and final Testing
4. **Cherish** - Working on UI & code for connection between server and RPI
5. **Ayush -** Soldering & working on code for connection between server and RPI

TOWARDS FINAL DEMO

1. We are waiting for the PCB to come and checking all the things end to end using PCB
2. UI Design is to be worked upon
3. Soldering Components on PCB
4. Final testing on PCB using 3-phase Load and debugging (if required)

THANK YOU